EVSP DR Current Research

Week of 14 September. Results checked 12 September 2026 at 15:34 EDT; overnight extensions are still running. Open Figures with explanations for timing, charging costs and geography, or CG curves and bus schedules for the restored figure library.

Current finding. With set covering and inherited columns, all six chains match the GIRO fleet at k=3–6. Chains 3, 5 and 6 also reach ten buses at k=10. The new 512-route import treatment reaches 11 buses at k=11 in chain 6; other completed extension cases need extra buses.

Integer fleet with full-pool inheritance

Each cell is the number of integer bus routes found. Green matches the target. Red requires extra buses. Dashes denote interrupted CG; see the C2 k10 note below.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
2	2	2	2	2	3	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	—	7	7	7	7	7
8	—	8	8	8	8	8
9	—	—	9	9	9	9
10	—	—	10	—	10	10

The table above retains the earlier full-pool treatment. The extension checks at most 512 inherited routes before CG; its new integer results are shown separately below. Runs continue through k=15 for all six chains.

k=2 starts from single-trip routes; later sizes reuse validated previous-k columns. Chain 3 k=10 also includes routes from a fresh solver solution. These are pool proofs. Chain 2 k10 (71 buses) had zero pricing iterations and no CG certificate. Individual route replay passed; duplicate-trip removal and shared charger capacity are not established by this table.

Fresh starts on the same chains

Each size starts independently. Both tables use covering and 240-kWh batteries with 240-kW charging. Fresh fleet proof status varies by cell.

Target buses	Chain 1	Chain 2	Chain 3	Chain 4	Chain 5	Chain 6
5	5	5	5	5	6	5
8	9	9	9	9	9	8
10	11	11	11	11	11	11
15	18	17	18	19	16	19

Inherited columns improve the available integer pool: chain 3 at k=8 improves from a proved nine-bus pool optimum to eight; chain 5 at k=5 improves from a proved six-bus pool optimum to five.

What is limiting progress

Checking saved routes took too long. Before starting CG on a larger trip set, we check that routes saved from the previous size are still valid. In chain 1 at k=7, chain 2 at k=9 and chain 4 at k=10, this checking used up the available time. We now check at most 512 saved routes, for at most 15 minutes, then let CG search for new routes. All three cases have now finished CG. Their integer solves found 8, 10 and 11 buses, respectively; only the first and third are proved minimums within their saved pools. Using fewer saved routes may affect the final integer solution, so this is a separate experiment.

Charger limits made the search for new routes very slow. In one three-bus test, the LP solved in 0.006 seconds, but one search for a new route took 7.15 hours. The run ran out of time before it could prove that no improving route remained. Its 16-bus integer solution uses only the routes found so far; it does not show that 16 buses are necessary. The timing figures explain these two different delays.

Case	Baseline	PARX 60 kW	Station capacity	Both
k=1, two duties tested	1 each	1 each	1 each	1 each
k=2, 23 trips	2	2	3	3
k=3, 35 trips	3	3	16	16

Sixteen is the result in a small generated pool, not a proved physical requirement. Capacity-constrained schedules passed station sweeps. This pilot has no 65% return-SOC floor and is separate from the six baseline chains.

Overnight extension results

16:48 EDT collection: 66/87 original CG certificates, plus the certified chain 2 k14 retry. There are now 69 MIP results under the original extension campaign (34 warm, 35 component), plus the recovered chain 2 k14 result described below. The table below retains the earlier 34 warm results; the new k14 result is 16 buses, pool bound 14, fleet unproved, CG 159.5 minutes, weighted LP 1,400,558.356.

Warm starts check ≤512 saved routes. All CG results below except chain 4 k=15 are pricing-certified. Times include preparation. Chain 4 k=15 has only a restricted LP objective, not a certified full-model bound.

Chain and target	Buses	Pool fleet bound	Fleet proved in pool	CG minutes	Weighted LP objective
Chain 1, k=07	8	8	Yes	53.6	700313.457
Chain 1, k=08	9	9	Yes	49.5	800383.688
Chain 1, k=09	10	9	No	67.9	900453.496
Chain 1, k=10	11	11	Yes	91.2	1000511.205
Chain 1, k=11	13	11	No	114.1	1100537.061
Chain 1, k=12	13	12	No	95.6	1200608.942
Chain 1, k=13	15	13	No	152.2	1300648.369
Chain 1, k=14	16	14	No	147.9	1400650.097
Chain 2, k=09	10	9	No	64.0	900301.353
Chain 2, k=10	11	10	No	55.7	1000371.146
Chain 2, k=11	12	11	No	65.7	1100404.250
Chain 2, k=12	13	12	No	85.6	1200441.362
Chain 2, k=13	15	13	No	98.4	1300493.195
Chain 3, k=11	12	11	No	27.7	1100375.859
Chain 3, k=12	14	12	No	32.8	1200427.901
Chain 3, k=13	14	13	No	52.6	1300474.559

Chain 3, k=14	15	14	No	57.9	1400485.597
Chain 3, k=15	17	15	No	59.7	1500507.420
Chain 4, k=10	11	11	Yes	42.0	1000426.271
Chain 4, k=11	12	11	No	59.4	1100462.865
Chain 4, k=12	13	12	No	68.2	1200505.601
Chain 4, k=13	15	13	No	88.6	1300530.332
Chain 4, k=14	18	14	No	204.6	1400590.723
Chain 4, k=15	18	15	No	239.1	1500637.979
Chain 5, k=11	12	11	No	70.5	1100522.228
Chain 5, k=12	13	12	No	109.2	1200576.644
Chain 5, k=13	14	13	No	134.6	1300625.367
Chain 5, k=14	16	14	No	140.1	1400639.061
Chain 5, k=15	17	15	No	209.7	1500669.159
Chain 6, k=11	11	11	Yes	32.5	1100399.029
Chain 6, k=12	13	12	No	64.6	1200464.548
Chain 6, k=13	15	13	No	117.1	1300532.041
Chain 6, k=14	16	14	No	136.9	1400578.400
Chain 6, k=15	18	15	No	176.3	1500606.338

Pool fleet bounds concern saved columns only. A proved fleet above target requires different columns to reach target. An unproved gap does not distinguish incomplete integer search from missing useful columns.

New results and remaining uncertainty

Chain 1 at k=14: CG finished with a pricing certificate after 147.9 minutes and 1,301 iterations. Its weighted LP objective is 1,400,650.097. The completed MIP found 16 buses, with a saved-pool fleet bound of 14; the fleet is unproved. Individual routes passed replay; duplicate removal and shared capacity remain unverified.

Chain 4 at k=15: the MIP found 18 buses with a saved-pool fleet bound of 15; neither fleet optimality nor full-model LP optimality is proved. CG had stopped at its time limit after 239.1 minutes, with last reduced cost −0.1998 against tolerance 0.0001. Individual routes passed replay; duplicate removal and shared capacity remain unverified.

Chain 2 at k=14, updated 12 September: CG reached a pricing certificate after 159.5 minutes and 1,484 iterations, with weighted LP objective 1,400,558.356. The subsequent one-hour MIP found **16 buses** and a **pool fleet bound of 14**; it did not prove the best fleet. Individual-route replay passed, but duplicate-trip removal and shared-station capacity were not validated. This is the earlier bounded512 warm start. The new full-pool chain is a separate experiment. The k15 CG continues from the recovered k14 routes, with its MIP queued afterward.

Decomposition of 32 duties

Grouping	Four component fleets	Combined buses	Target
join01	9 + 9 + 9 + 8	35	32
join02	9 + 9 + 9 + 8	35	32
join03	9 + 9 + 8 + 10	36	32
join08	9 + 9 + 10 + 9	37	32
join09	8 + 9 + 9 + 9	35	32

These saved constructions combine disjoint trip groups chosen using GIRO duty membership; no GIRO route columns were supplied. Component routes passed replay. Shared charger capacity and duplicate removal are not certified. **All five joined-parent attempts timed out before producing a parent CG result.** Their saved decomposed constructions remain available, but there is no joint improvement or parent optimality proof.

Completed speed comparisons

Case	Reference minutes	Optimized minutes	Observation
d00_g0	22.3	23.6	5.8% slower
d00_g1	52.2	52.1	Essentially unchanged
w1_k08	41.1	34.4	16.2% less time
w1_k08_repeat	31.1	24.1	22.4% less time

w4_k11	58.8	49.1	16.5% less time
w6_k12	67.2	54.6	18.6% less time

All six pairs reached matching certified LP objectives. Four warm comparisons used 16–22% less time; inherited-route checking fell from 366–447 seconds to 5–7 seconds. Common graph-cache preparation is excluded. CG iteration counts differed, so the total savings are descriptive, not an isolated causal estimate. Fresh comparisons show no speed improvement. No integer improvement is claimed from these timing tests.

Capacity comparisons at the time limit

Capacity case	Iterations ref / opt	Final restricted LP objective	Fractional route weight	Pricing certified ref / opt
cap_k1	13 / 12	100056.952000	1.0	No / No
cap_k1_combined_peak12	34 / 37	100074.693539	1.0	No / No
cap_k2	4 / 4	300094.976000	3.0	No / No

Every arm stopped after three hours. Both methods reached the same restricted-LP objective in each case; none proved that no improving route remained. Fractional route weight is not an integer fleet result. These tests do not demonstrate a convergence speedup.

Cluster queue — 12 September, 16:48 EDT

Work	Running	Waiting for input data
Full-pool warm chains through k=15	6	68
Decomposition: shared graph preparation	2	22
Earlier bounded warm chains	2	2

Recovered MIPs with ready CG outputs	4	0
Separate V2G project	9	0

We removed 43 obsolete queued entries and recovered nine MIPs whose data were already ready. Five have finished: each found 8 buses for its 8-bus subset, proved the fleet optimal within its saved column pool, and passed individual-route replay.

The remaining waits are genuine: the next k needs the previous k's routes; a MIP needs its CG output; the larger decomposition runs need their component solutions and shared graph. No dependency is marked impossible.

On Unicorn, run **~/ladder-lite/drq** for this grouped view, or add **--details** to see each prerequisite. Job map and recovery evidence.

Methods and evidence. CG route cost = 100,000 + electricity + 5 per charging start. MIP stage 1 minimizes buses. Stage 2 constrains buses ≤ the validated stage-1 incumbent and minimizes electricity plus start fees.

A pricing certificate establishes the LP result only in its represented graph and tolerance. A MIP proof concerns its saved columns. Fleet attainment, physical replay and shared-capacity validation are separate checks.

Experiment register · Dated results and source tables · Experiment settings · Historical document and figures · Local prototype tests and charging-cost bug

Storage, 12 September: 156 cold files archived losslessly; 133.74 GB freed. Active research and V2G data preserved. Restore instructions.

Blocked decomposition case d00_g3: the original run timed out before CG, while constructing the graph. Its MIP remains blocked. A separate diagnostic crashed after 366 seconds (SIGSEGV), before its watchdog limit. Partial samples show JSON serialization while adding charging arcs; A local timer-only test also hung, so the asynchronous sampler has been retired. The exact cluster crash remains unresolved. Diagnostic evidence. No long retry has been launched.